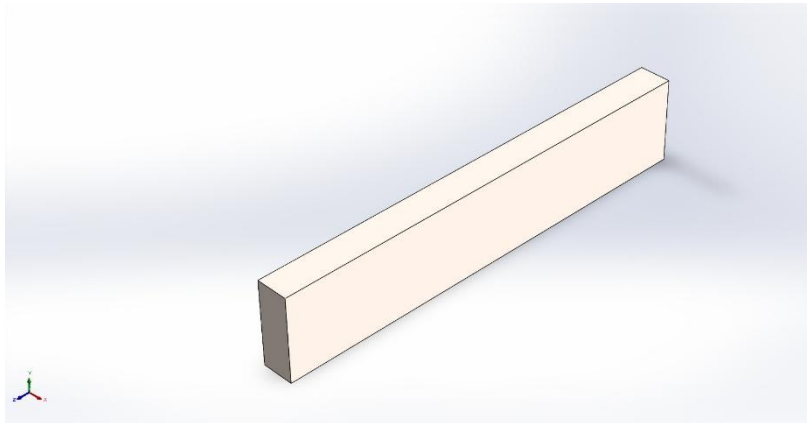




Simulation of beam load analysis

Date: Friday, February 13, 2026

Designer: MD Naiem Gazi

Study name:

DualPlane_Loaded_Beam_OC_Uniform_Point_Load

Analysis type: Static

Table of Contents

Description	1
Assumptions	2
Model Information	2
Study Properties	3
Units	3
Material Properties	4
Loads and Fixtures	4
Connector Definitions	5
Interaction Information	5
Mesh information	5
Sensor Details	5
Resultant Forces	6
Beams	6
Study Results	7
Conclusion	8

Description

1. Geometry of the Cross-Section

Shape: Rectangular (as shown in the figure).

Dimensions:

Width (in the z-direction): 1.5 in

Height (in the y-direction): 0.75 in

Length of the Beam: 8 in

2. Alternative Cross-Section (Part b)

Shape: Solid circular rod.

Diameter (d): 1.25 in

3. Material Properties : Alloy Steel. 4. Loading Conditions

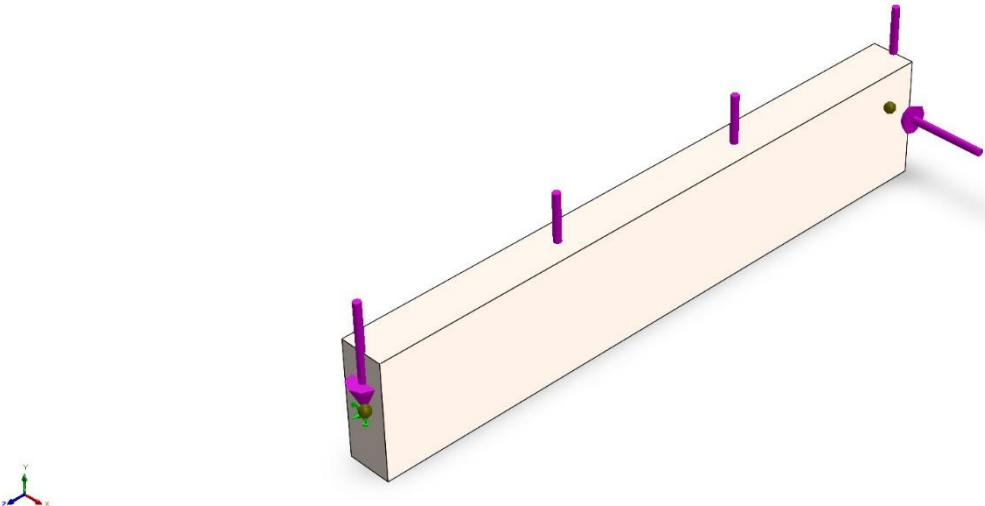
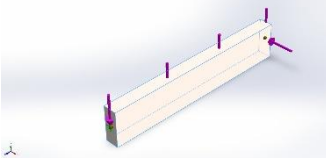
Uniform Load (xy-plane): 50 lbf/in (acting downward in the y-direction).

Concentrated Force (xz-plane): 100 lbf (acting downward in the z-direction at end C).



Assumptions

Model Information

 Model name: beam load analysis Current Configuration: Default			
Beam Bodies:			
Document Name and Reference	Formulation	Properties	Document Path/Date Modified
SolidBody 1(Boss-Extrude1) 	Beam - Uniform C/S	Section Standard-Custom Section Area: 0.000557647m^2 Length:203.2mm Volume: 0.000113314m^3 Mass Density: $7,700\text{kg/m}^3$ Mass:0.872516kg Weight:8.55066N	E:\L-3 T- 2\Assignments\Part 1\beam load analysis.SLDPRT Jan 29 13:08:09 2023

Study Properties

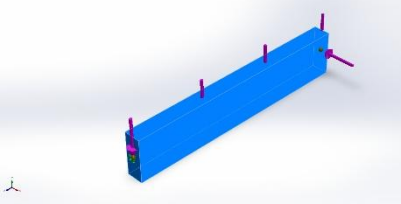
Study name	DualPlane_Loaded_Beam_OC_Uniform_Point_Load
Analysis type	Static
Mesh type	Beam Mesh
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Result folder	SOLIDWORKS document (E:\L-3 T-2\Assignments\Part 1)

Units

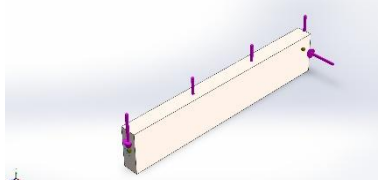
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

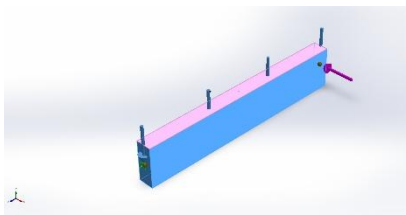
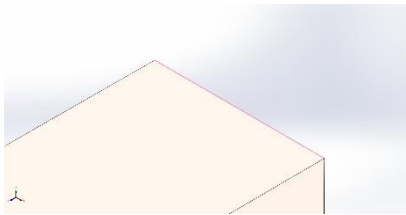


Material Properties

Model Reference	Properties	Components
	Name: Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m ² Tensile strength: 7.23826e+08 N/m ² Elastic modulus: 2.1e+11 N/m ² Poisson's ratio: 0.28 Mass density: 7,700 kg/m ³ Shear modulus: 7.9e+10 N/m ² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Boss-Extrude1)(beam load analysis)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 Joint(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-1		Entities: 1 Beam (s) Reference: Face< 1 > Type: Apply force Values: ---, ---, 50 lbf/in Moments: ---, ---, --- lbf.in/in
Force-2		Entities: 1 Joint(s) Reference: Edge< 1 > Type: Apply force Values: ---, ---, 100 lbf Moments: ---, ---, --- lbf.in

Connector Definitions

No Data

Interaction Information

No Data

Mesh information

Mesh type	Beam Mesh
-----------	-----------

Mesh information - Details

Total Nodes	44
Total Elements	42
Time to complete mesh(hh:mm:ss):	00:00:01
Computer name:	HELAL

Sensor Details

No Data



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	444.822	1,779.29	4.47722e-32	1,834.05

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	180.776	-90.3879	1.40171e-22	202.113

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0	0	0	0

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Beams

Beam Forces

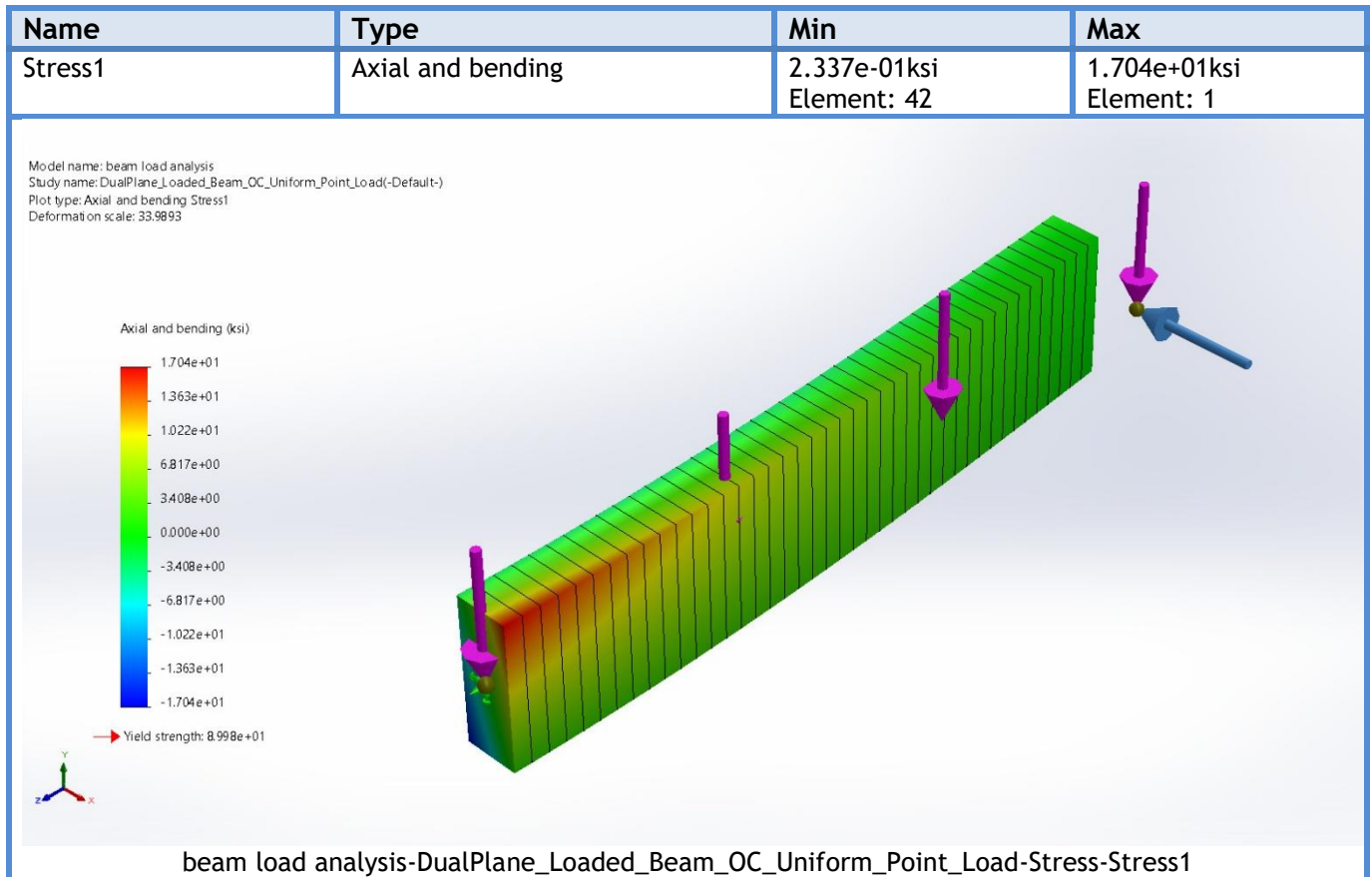
Beam Name	Joints	Axial(N)	Shear1(N)	Shear2(N)	Moment1(N.m)	Moment2(N.m)	Torque(N.m)
Beam-1(Boss-Extrude1)	1	4.47722e-32	1,779.29	444.822	-90.3879	180.776	-1.40171e-22
	2	0	-2.32322e-07	-444.822	-4.49652e-11	5.74476e-10	0

Beam Stresses

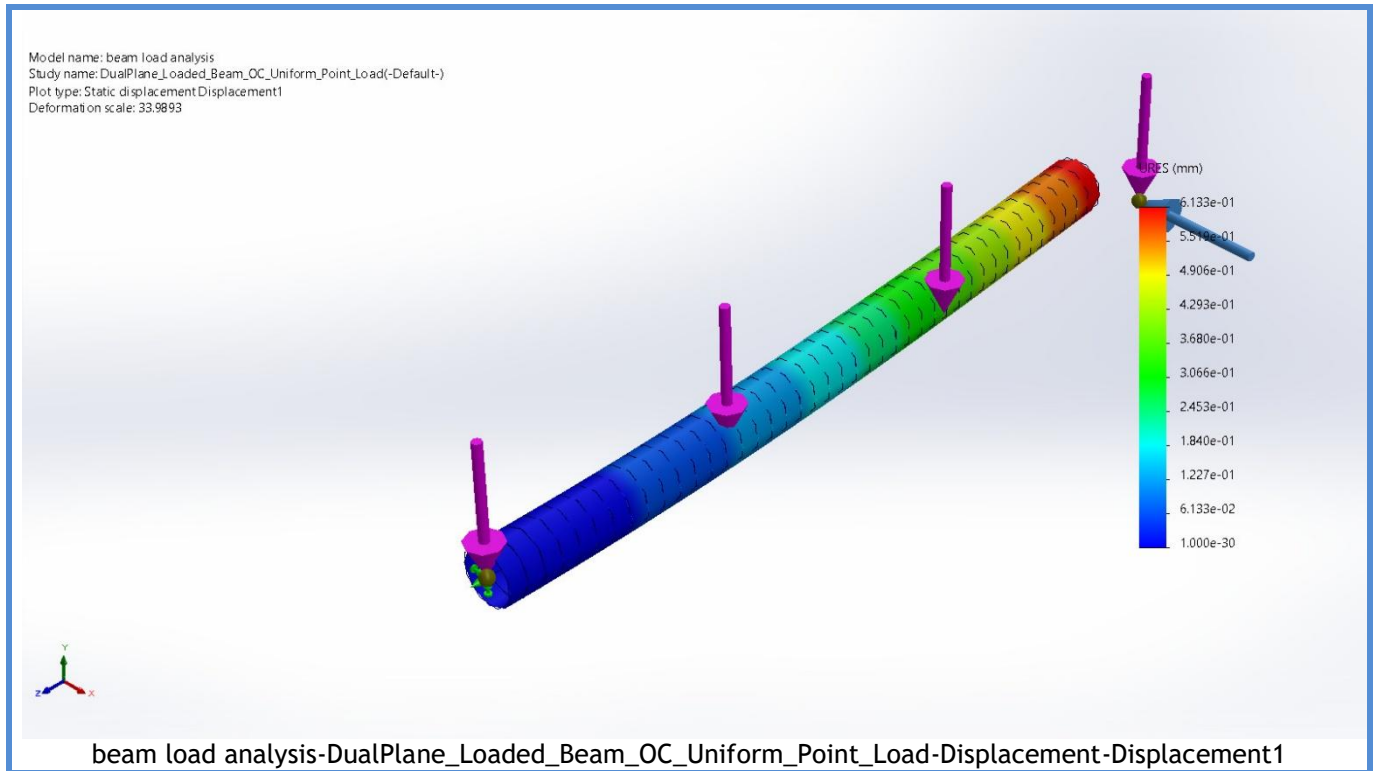
Beam Name	Joints	Axial(N/m ²)	Bending Dir1(N/m ²)	Bending Dir2(N/m ²)	Torsional (N/m ²)	Upper bound axial and bending(N/m ²)
Beam-1(Boss-Extrude1)	1	8.02878e-29	6.64459e+07	5.10514e+07	9.47045e-17	1.17497e+08
	2	-1.5018e-32	-3.30548e-05	-0.000162233	3.36823e-29	0.000195288



Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 1	6.133e-01mm Node: 43



Conclusion

